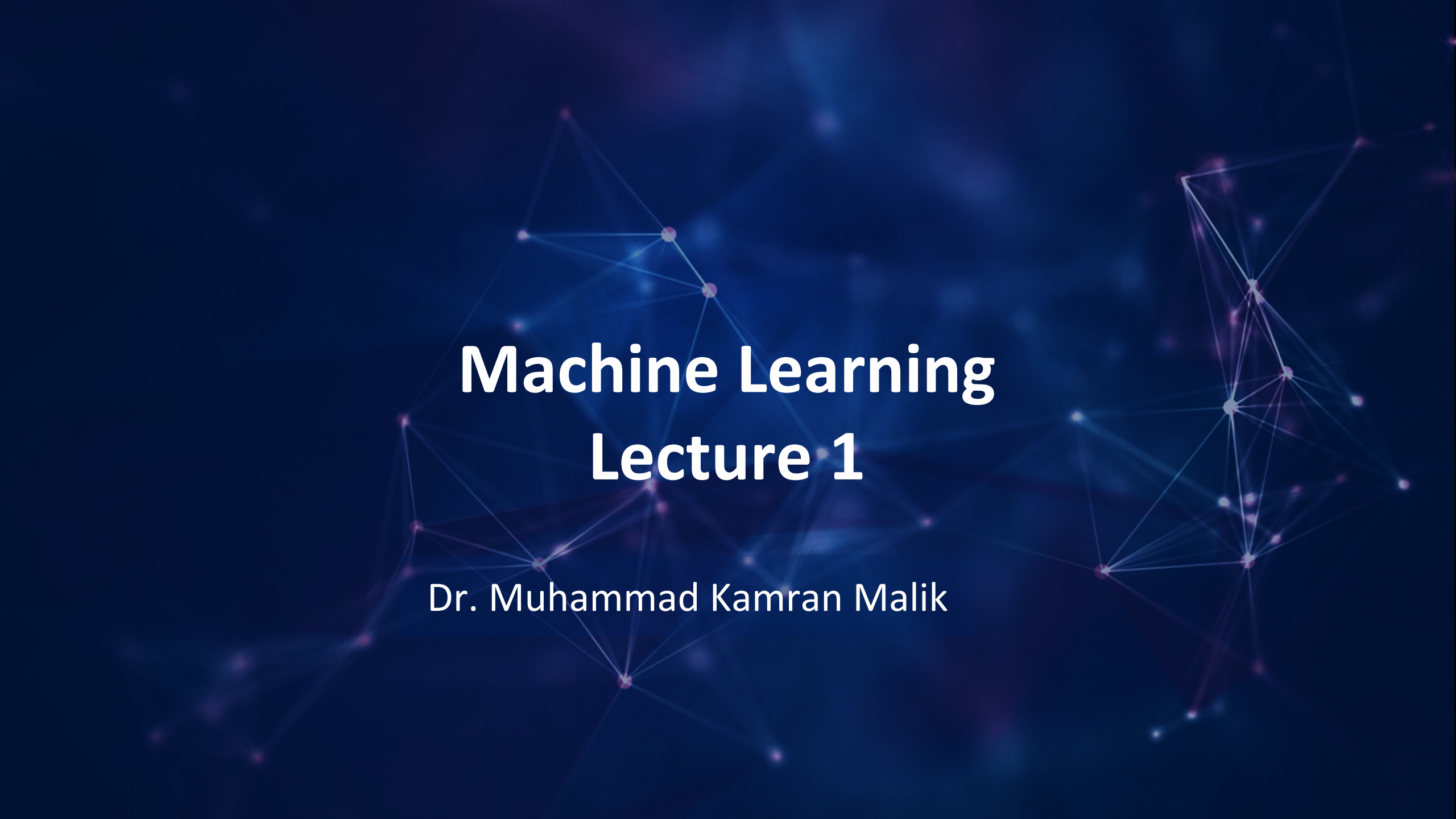




Machine Learning Lecture 1

Dr. Muhammad Kamran Malik

Assignments

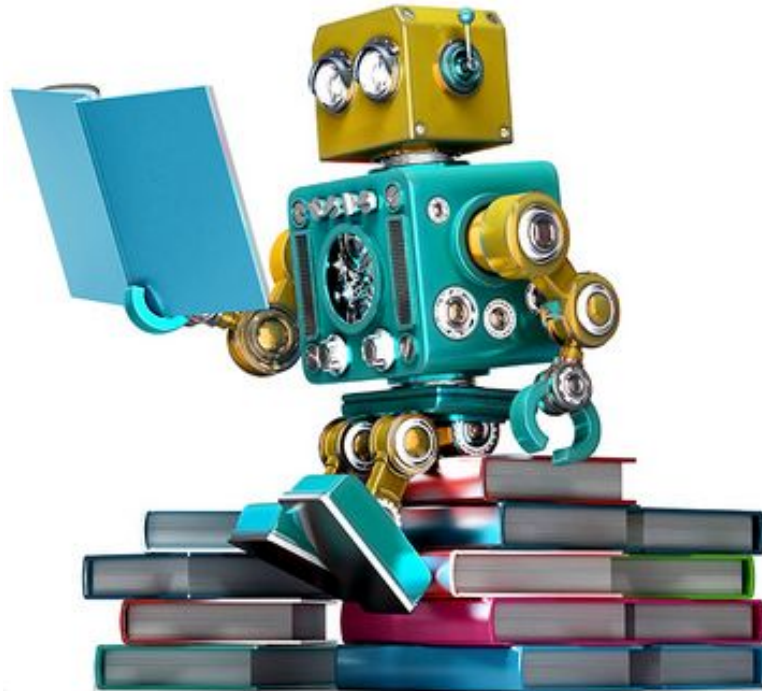
- Video Search Engine (searching a dialog in video/audio and finding its occurrences)
- Urdu to English and vice versa (Speech Translation System)
- Python (especially list comprehension, dictionary comprehension etc and numpy)



What is Machine Learning

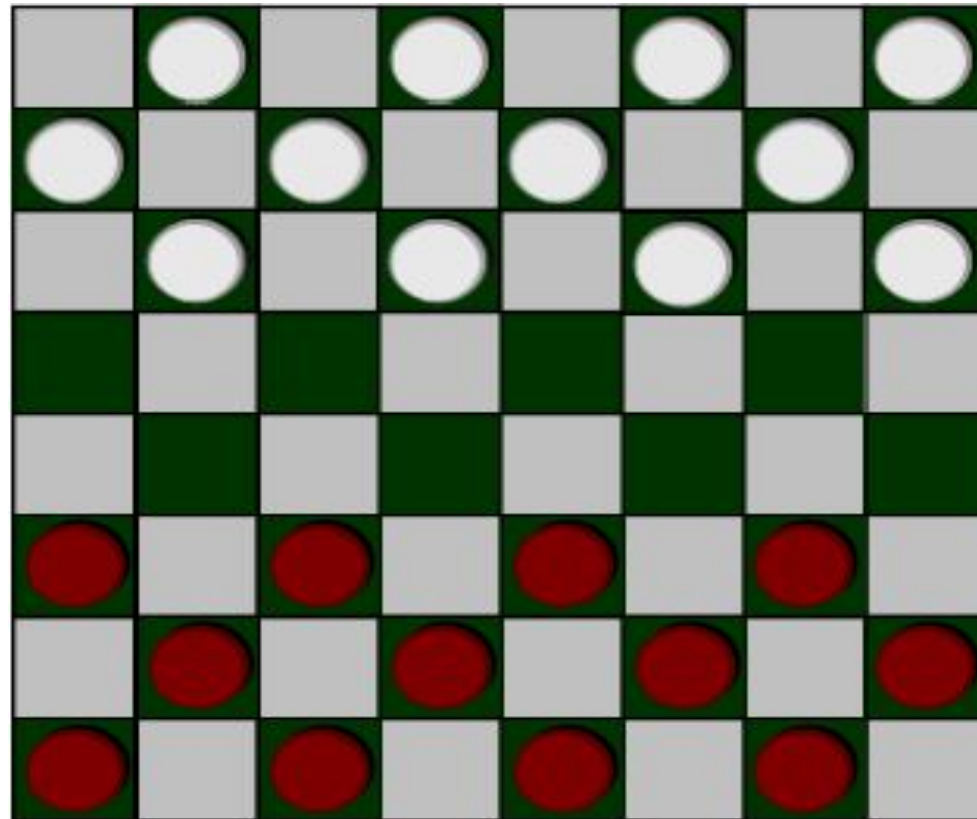
What is Machine Learning

- A subfield of computer science and AI
- Learn, make decisions and prediction from data.
- Enables computers to act and make data-driven decision rather than being explicitly programmed to carry out a certain task.



What is Machine Learning

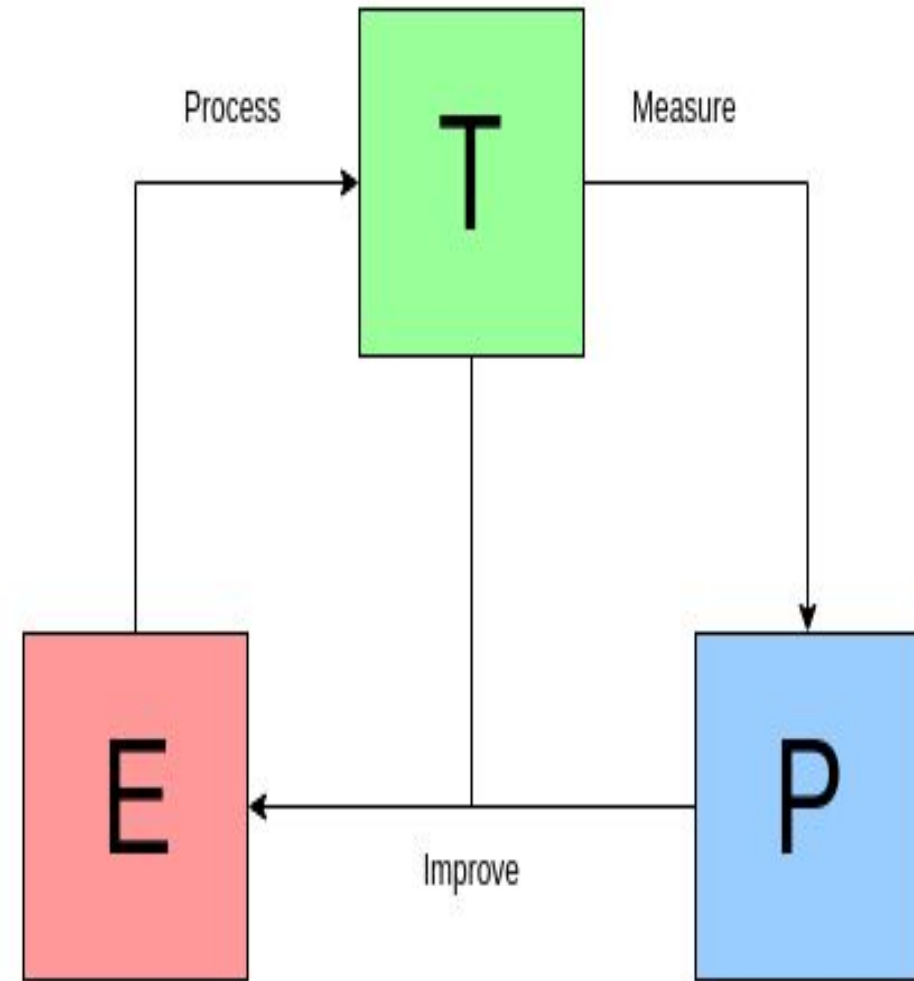
- The field of study that gives computers the ability to learn without being explicitly programmed. (Arthur Samuel-1959)



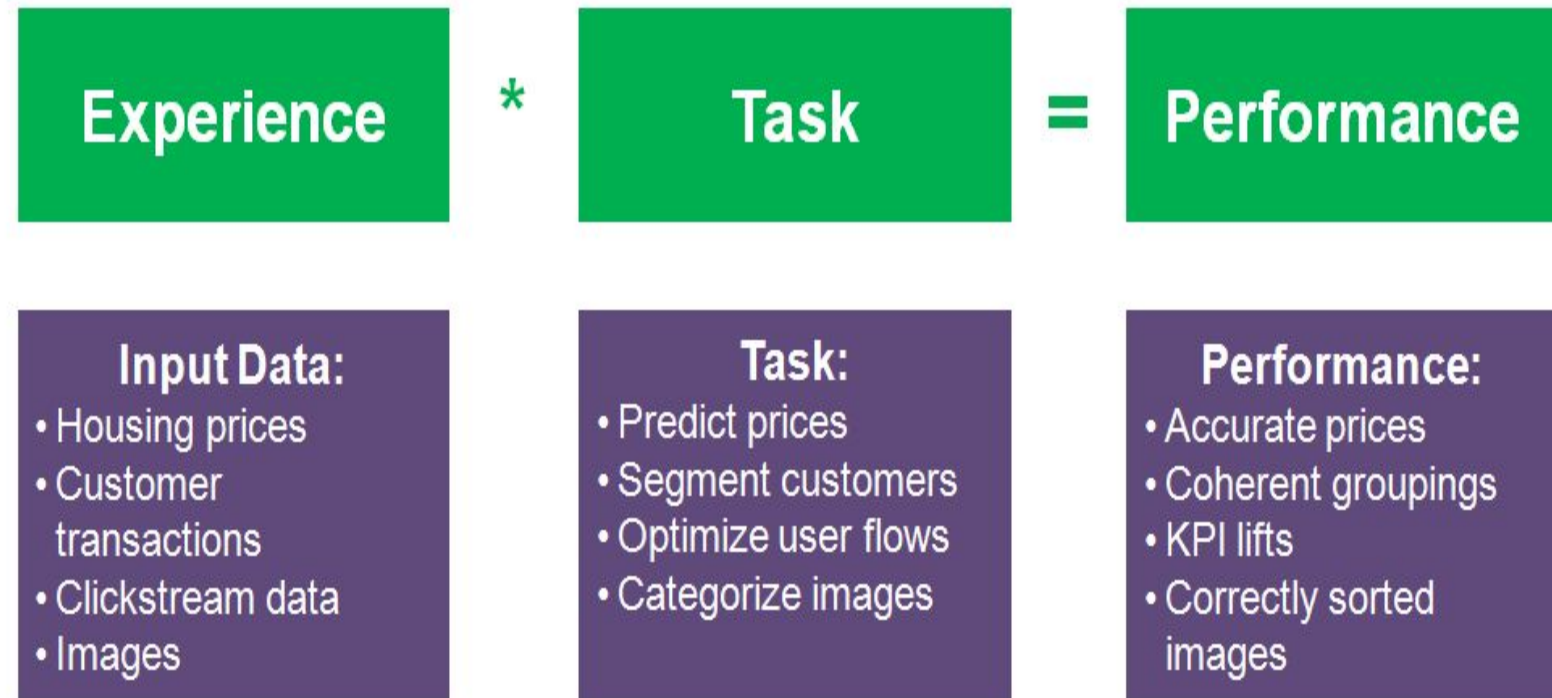
What is Machine Learning

- A computer program is said to learn from experience E with respect to some class of tasks T and performance measure P , if its performance at tasks in T , as measured by P , improves with experience E . (Tom Mitchell-1998)
- Playing checkers
 - E = the experience of playing many games of checkers
 - T = the task of playing checkers.
 - P = the probability that the program will win the next game.

What is Machine Learning



$$E * T = P$$



Example: Email Spam Checker

- "A computer program is said to learn from experience E with respect to some task T and some performance measure P , if its performance on T , as measured by P , improves with experience E ."
- Suppose your email program watches which emails you do or do not mark as spam, and based on that learns how to better filter spam. What is the T , E and P in this setting?



Gmail



Compose



Chats



Scheduled



All Mail



Spam

87



Trash



Manage labels



Dr. Muhammad



Example: Email Spam Checker

- "A computer program is said to learn from experience E with respect to some task T and some performance measure P , if its performance on T , as measured by P , improves with experience E ."
- Suppose your email program watches which emails you do or do not mark as spam, and based on that learns how to better filter spam. What is the task T in this setting?
 - Classify emails as spam or not spam.
 - Watching you label emails as spam or not spam.
 - The number (or fraction) of emails correctly classified as spam/not spam.
 - None of the above, this is not a machine learning algorithm.

When Machine Learning?

- Learning is used when:
 - Human expertise does not exist (navigating on Mars),
 - Humans are unable to explain their expertise (speech recognition, time series analysis)
 - Solution changes in time (routing on a computer network)
 - Solution needs to be adapted to particular cases (user biometrics)
 - Data is Big Data (Volume, Velocity, Variety)



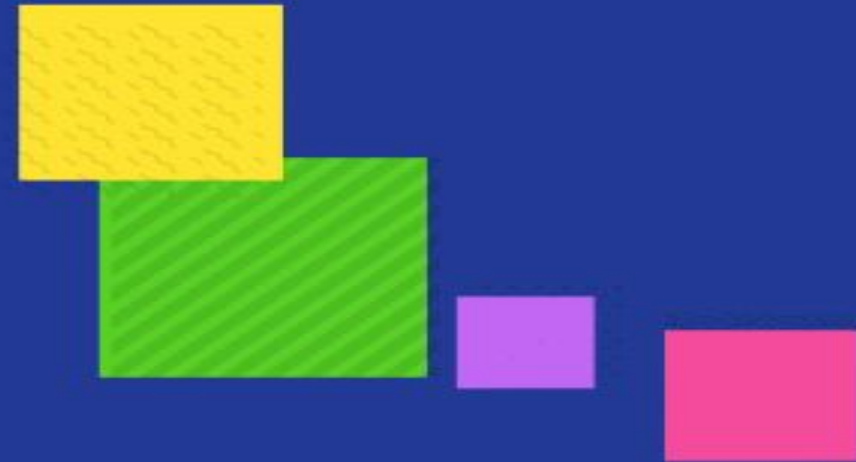
Q & A



Machine Learning Applications

Machine Learning Applications (Virtual Personal Assistant)

- Calendar & Meeting Reminder
- Automation
- Natural Conversation
- Recommending
- Smarter Learning
- Integration

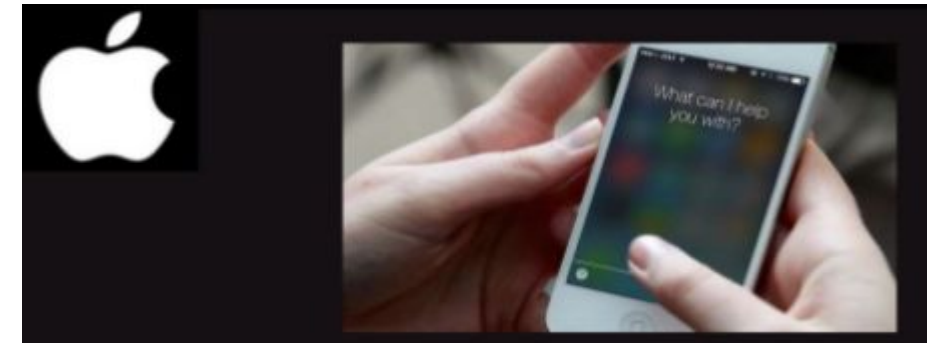
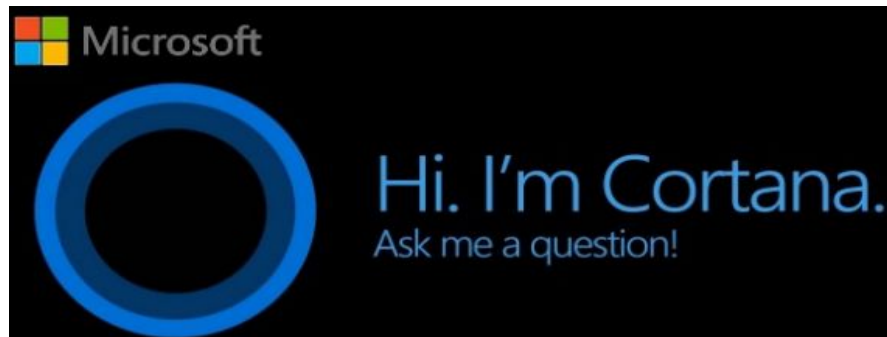


Machine Learning Applications (Virtual Personal Assistant)

Google Assistant



Hi, how can I help?



SILVIA



Viv



Bixby

rai:

Braina

'CR

Mycroft

Machine Learning Applications (Traffic Prediction)



Google Maps predicts whether the traffic is **clear**, **slow-moving** or **heavily congested** based on two measures:

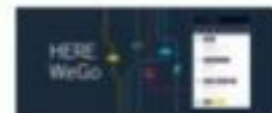
- ❑ **Average time taken** on specific days at specific times on that route
- ❑ **real time location** data of vehicles from **google maps** application and **Sensors**

MOST POPULAR MAP SERVICES



 **Bing maps**

 **MAPS.ME**



Machine Learning Applications (Email Spam Filtering)



Gmail is one of the many popular email providers who have an inbuilt spam filter:

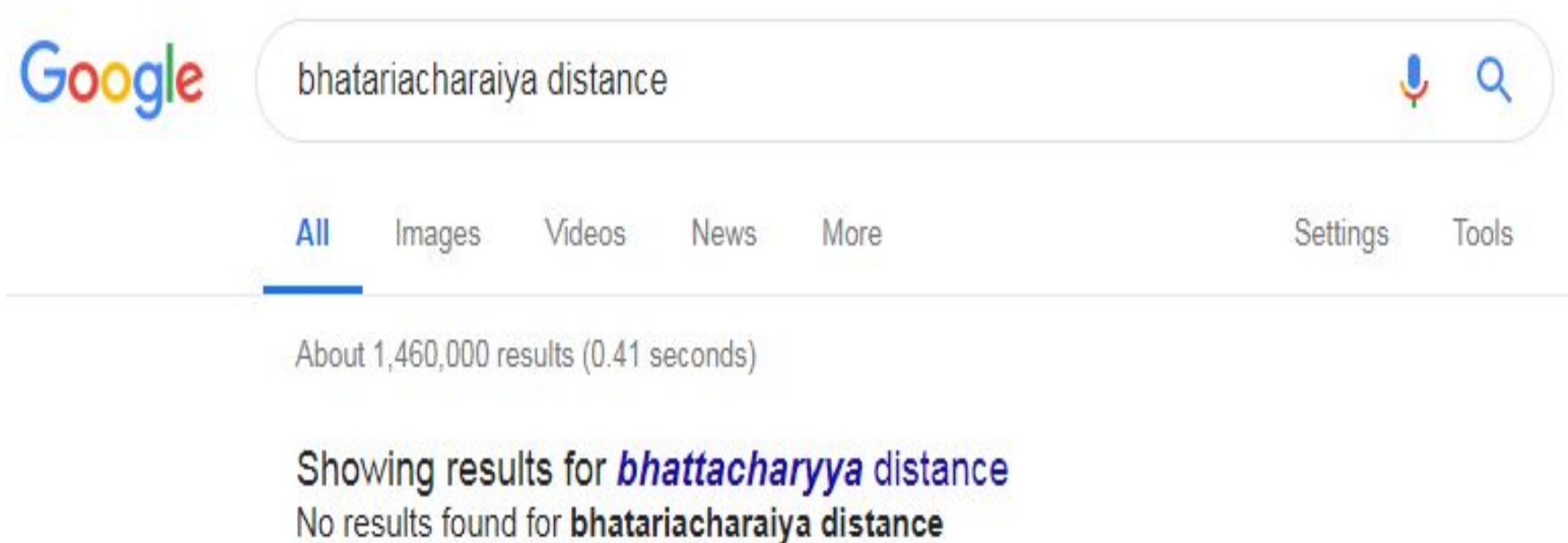
Spam filters are of the following types:

- ☐ Content filters
- ☐ Header filters
- ☐ General blacklist filters
- ☐ Rules-based filters
- ☐ Permission filters
- ☐ Challenge-response filters

Machine Learning Applications (chatbots)



Machine Learning Applications (Search Engine Result Refining)



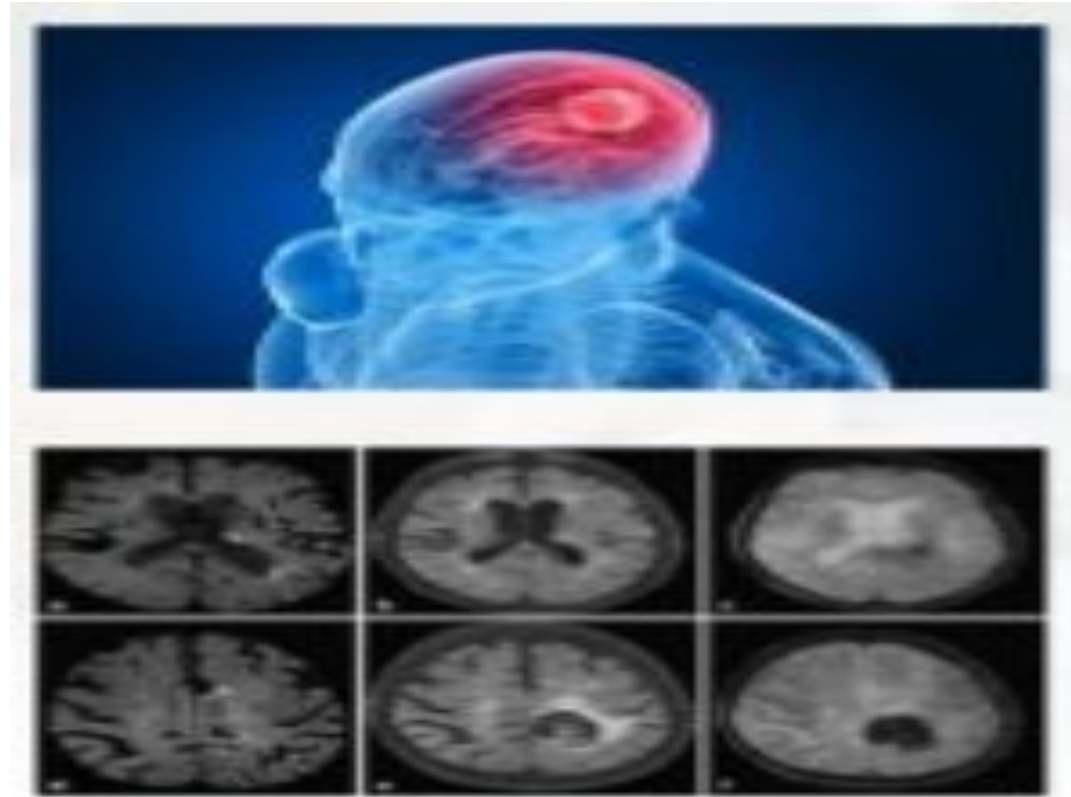
Machine Learning Applications (Online Fraud Detection)



Machine Learning Applications (Stock Market Trading)



Machine Learning Applications (Assistive Medical Technology)



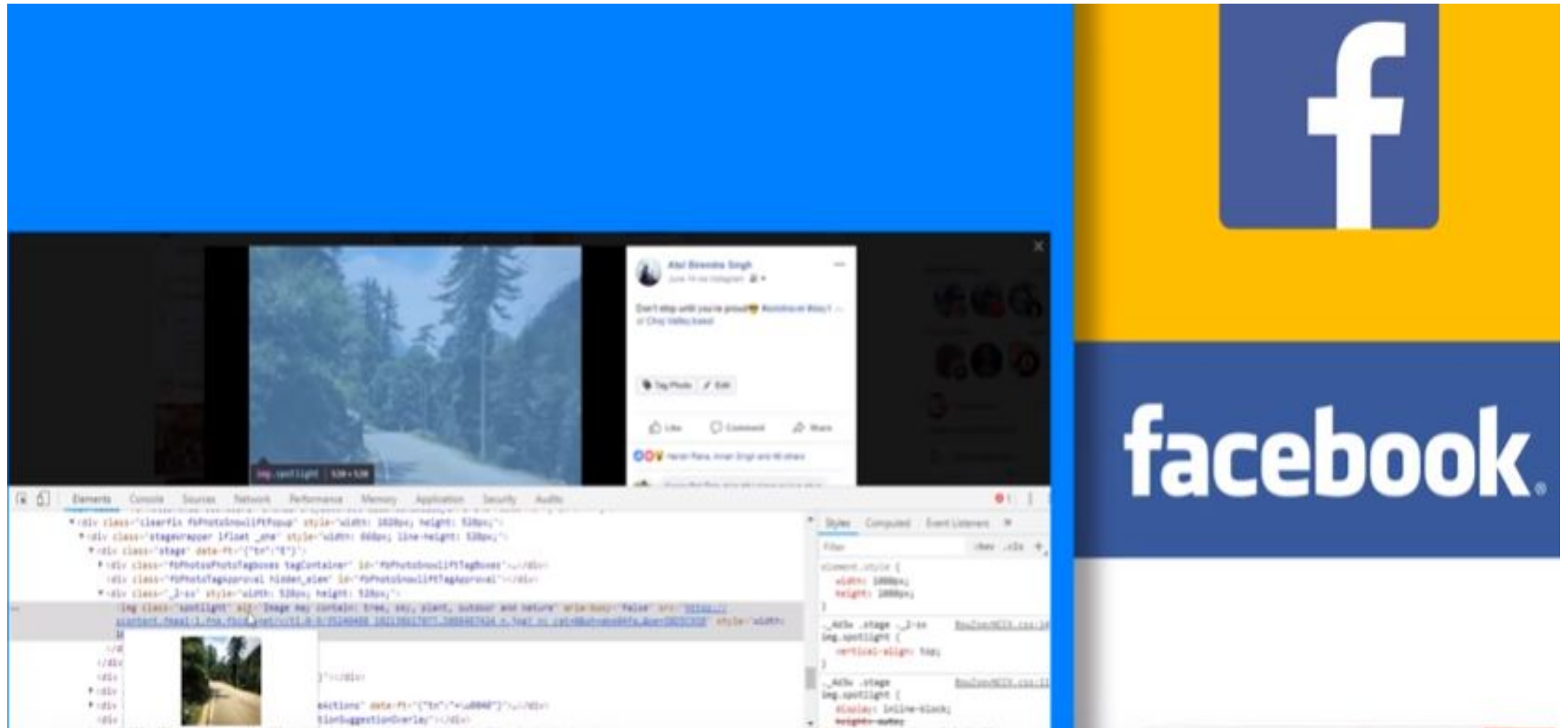
Machine Learning Applications (Automatic Translation)



Machine Learning Applications (Automatic Friend Tagging suggestions)



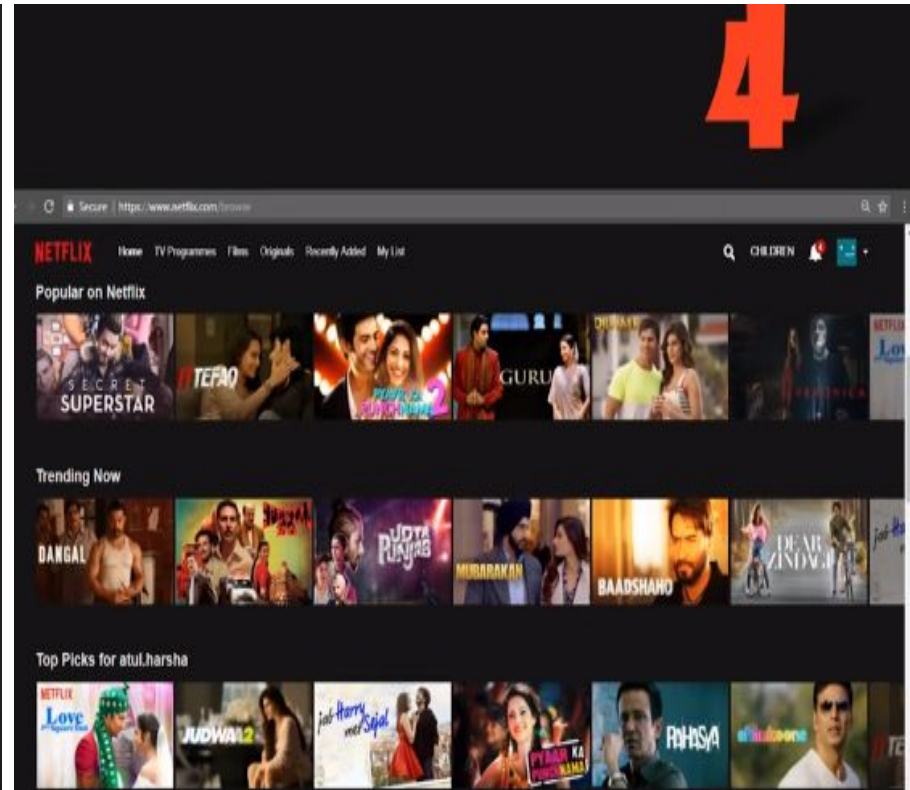
Machine Learning Applications (Alt Tags)



Machine Learning Applications (Recommendation Systems)



Ads Recommendation



NETFLIX

Recommender System

Machine Learning Applications

- Autonomous Cars (TESLA)
- Face Recognition app (Iphone)
- Moley (The Robotic Chef)
- KUKA (Robotic Automation)

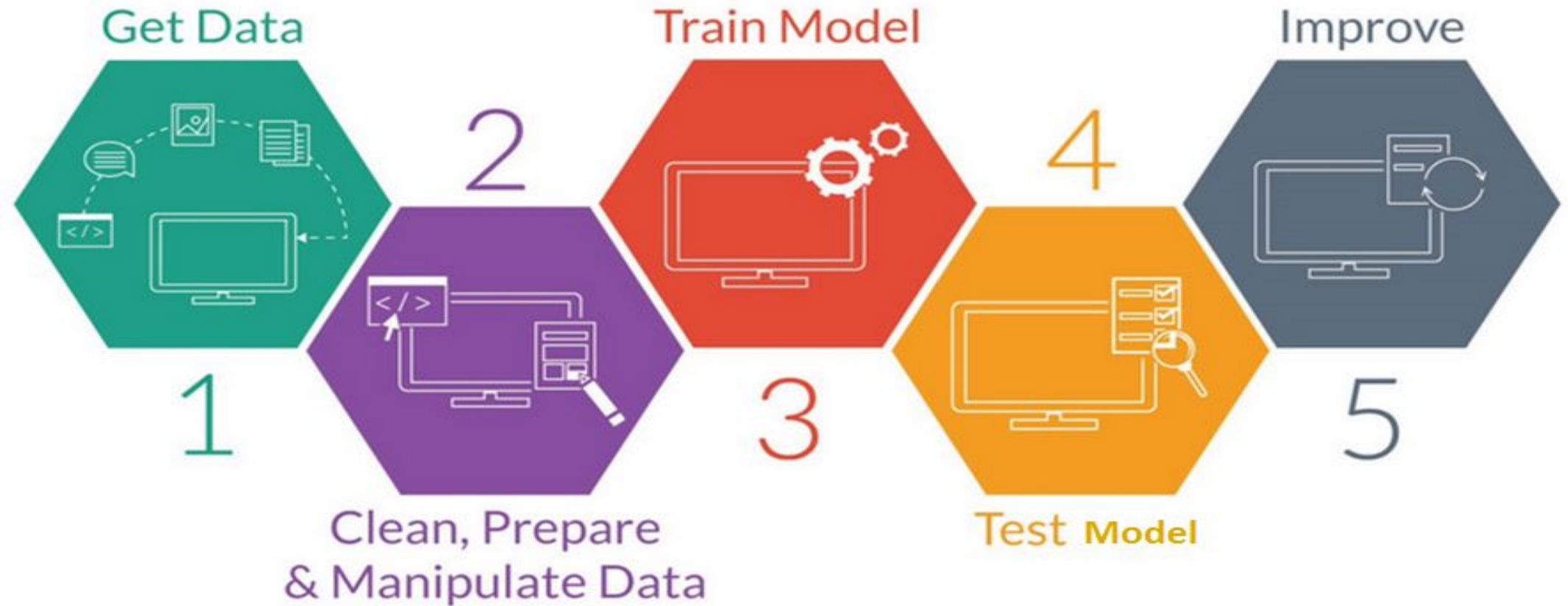


Q & A

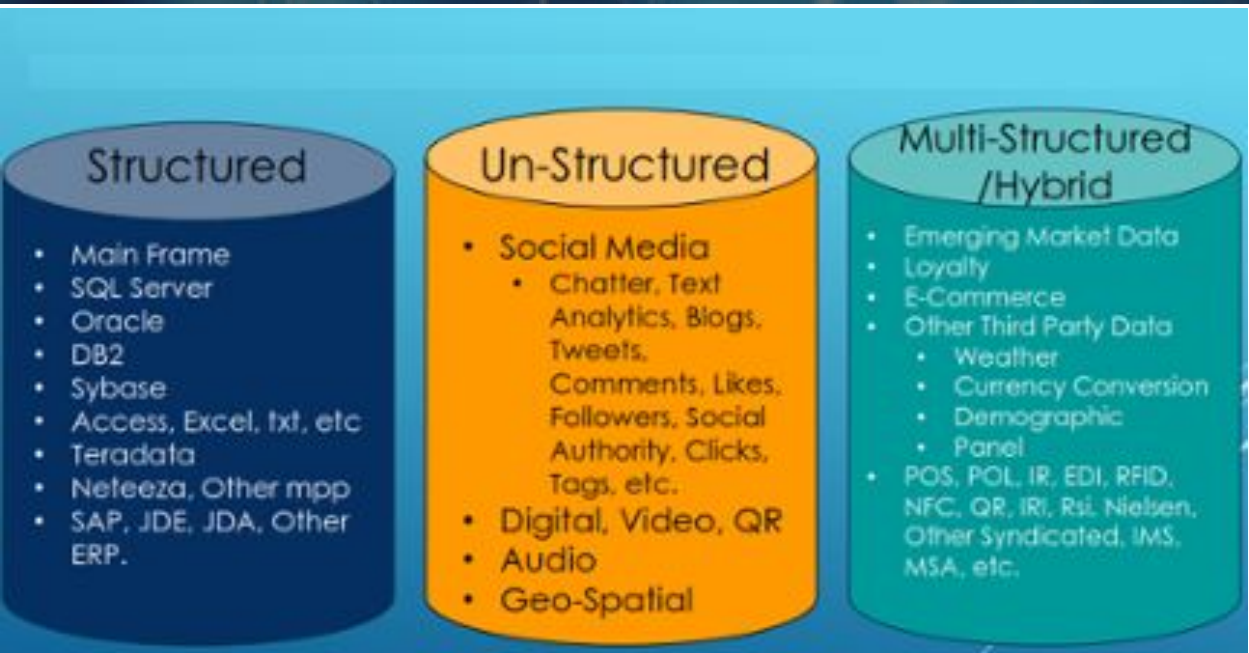


Machine Learning Process Lifecycle

Machine Learning Process Lifecycle



Get Data



Sound



Number/Categorical

Age	Height	Weight	Gender
19	160	55	male
25	170	55	male
30	175	60	femal

Text



Image



Cleaning and preparing (Features)

TRADITIONAL MACHINE PERCEPTION - HAND TUNED FEATURES

Raw data



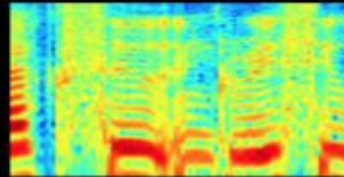
Feature extraction



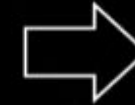
Classifier/
detector

SVM,
shallow neural net,
...

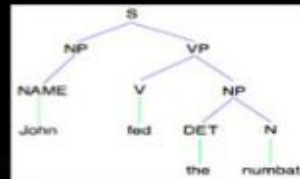
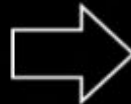
Result



HMM,
shallow neural net,
...



Speaker ID,
speech transcription, ...



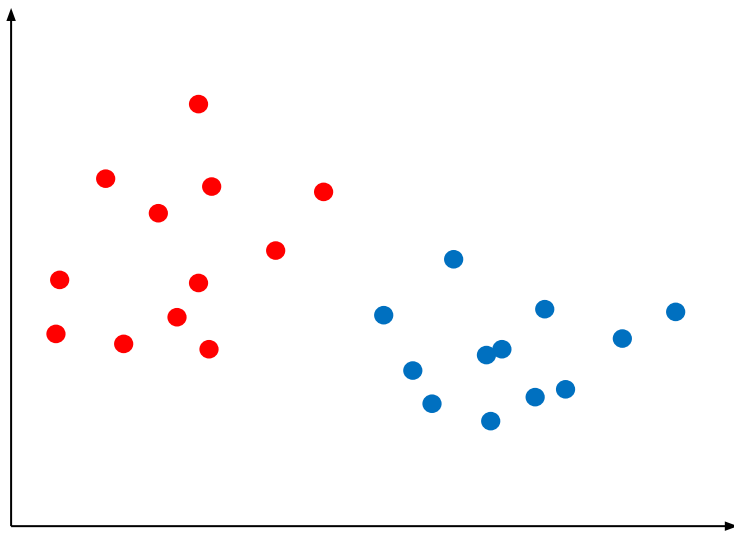
Clustering, HMM,
LDA, LSA
...



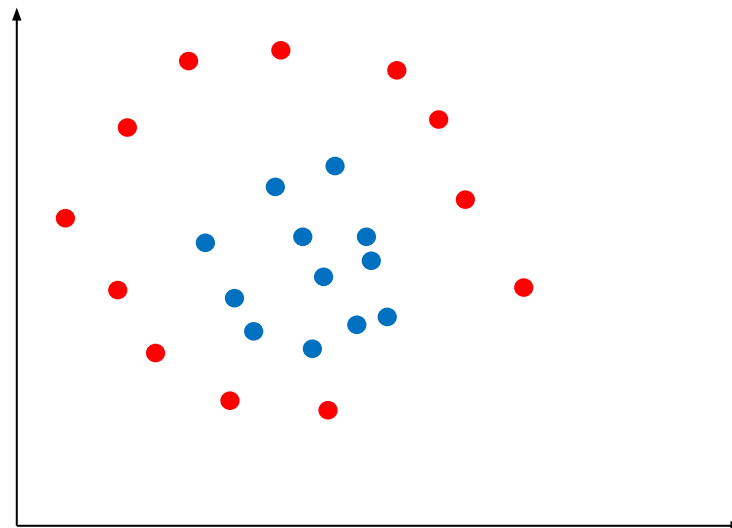
Topic classification,
machine translation,
sentiment analysis...

Train Model (Model Building – a search problem)

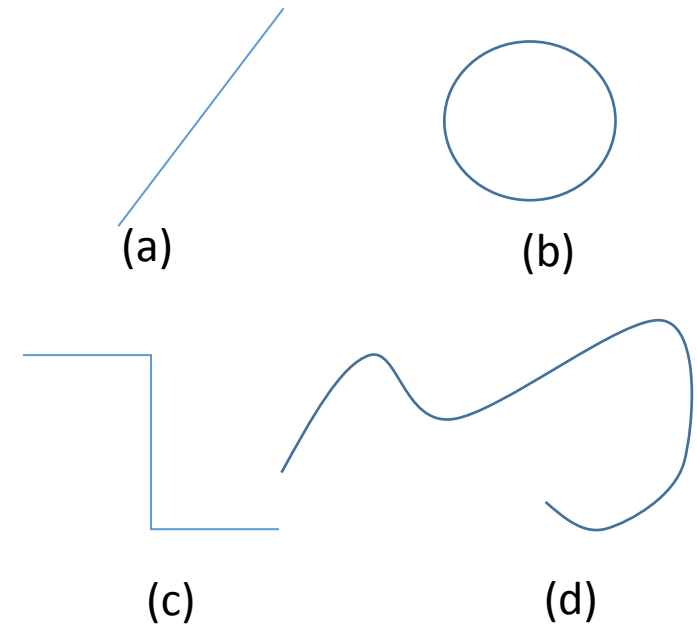
- Which type of model is most suitable for this data?



Dataset - 1



Dataset - 2



Test Model (Evaluation)

- Given prediction results, which model is better?

Model-1	Predicted		
Actual			
	100 	50	50
	1000 	100	900

Accuracy = $950 / 1100 = 86.4\%$

Model-2	Predicted		
Actual			
	100 	0	100
	1000 	0	1000

Accuracy = $1000 / 1100 = 91\%$

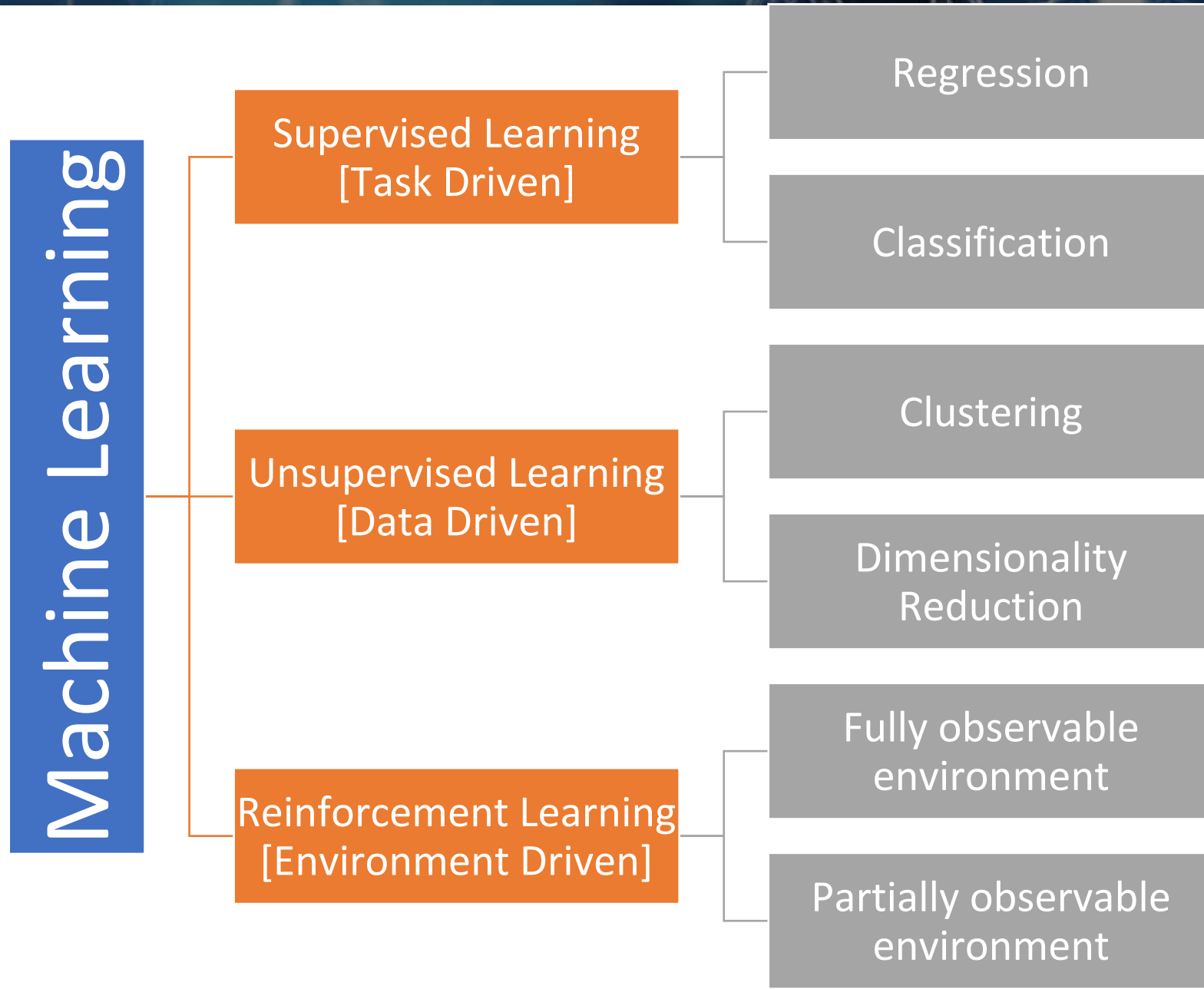


Q & A

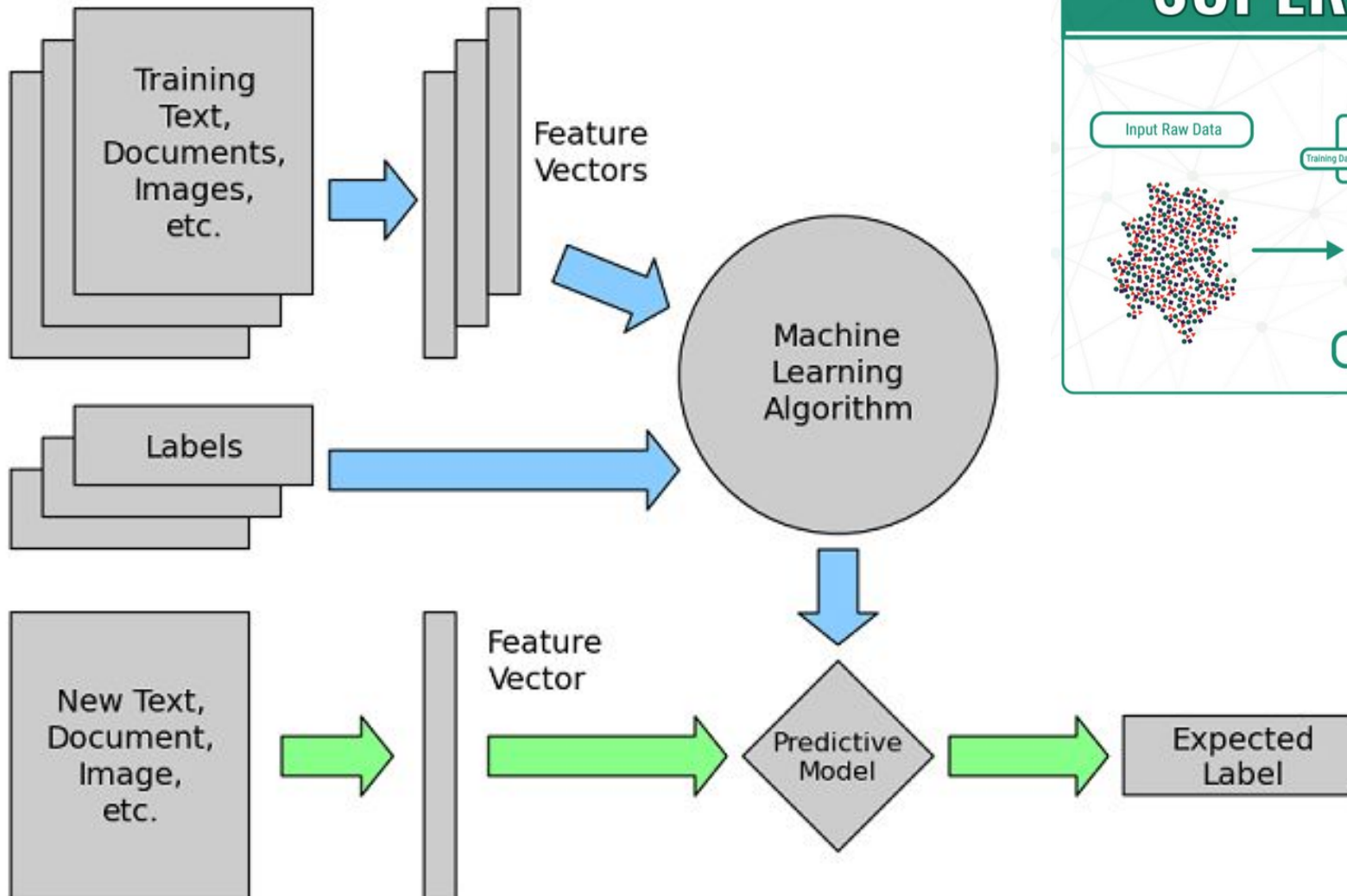


Types of Machine Learning Algo

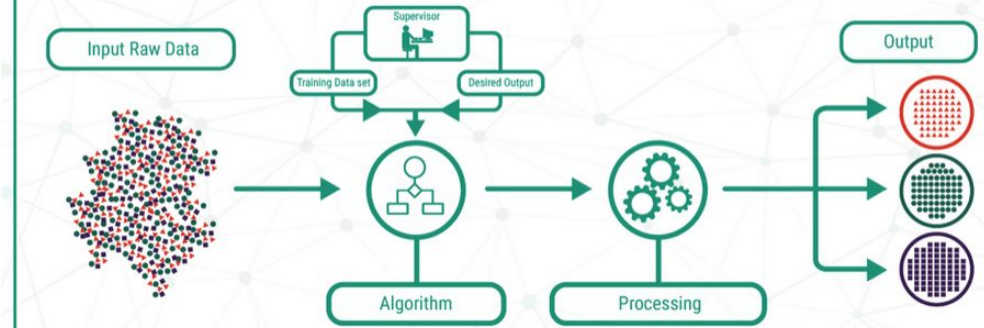
Types of Machine Learning Algorithms



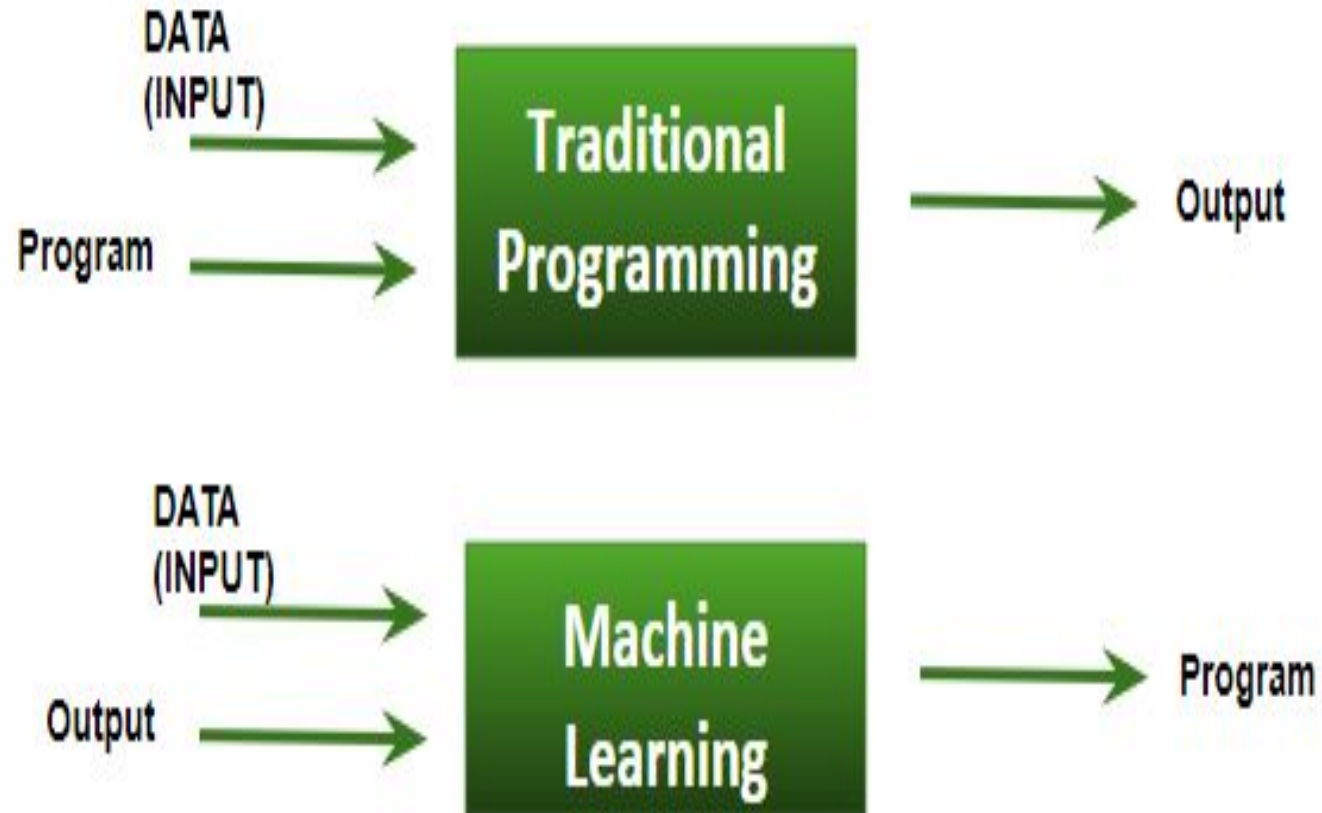
Supervised Learning



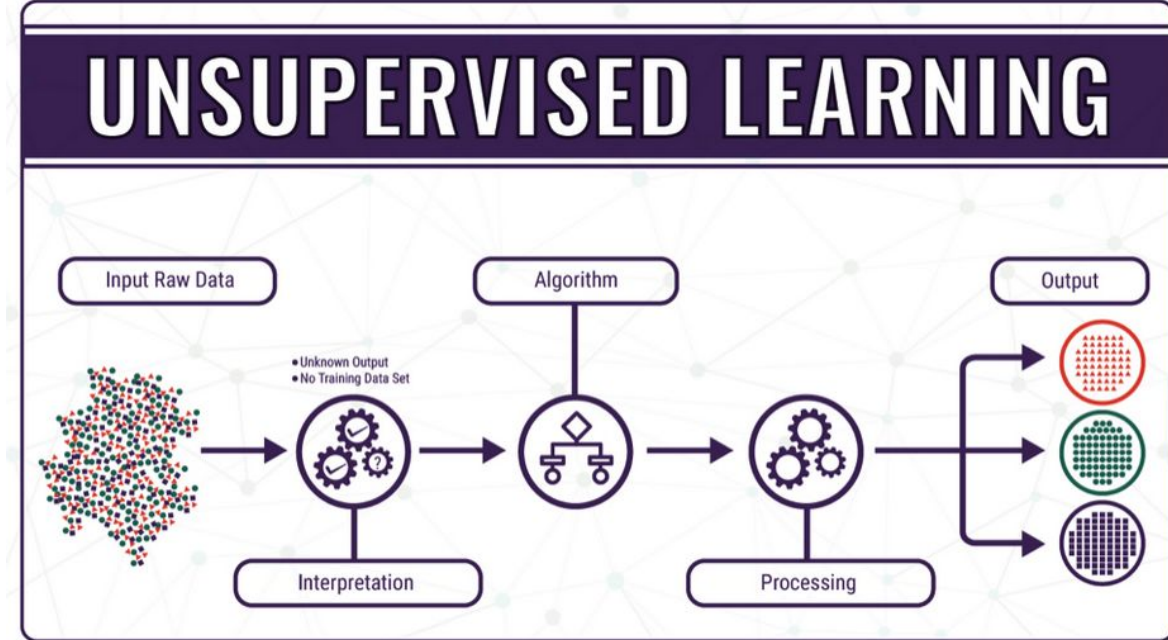
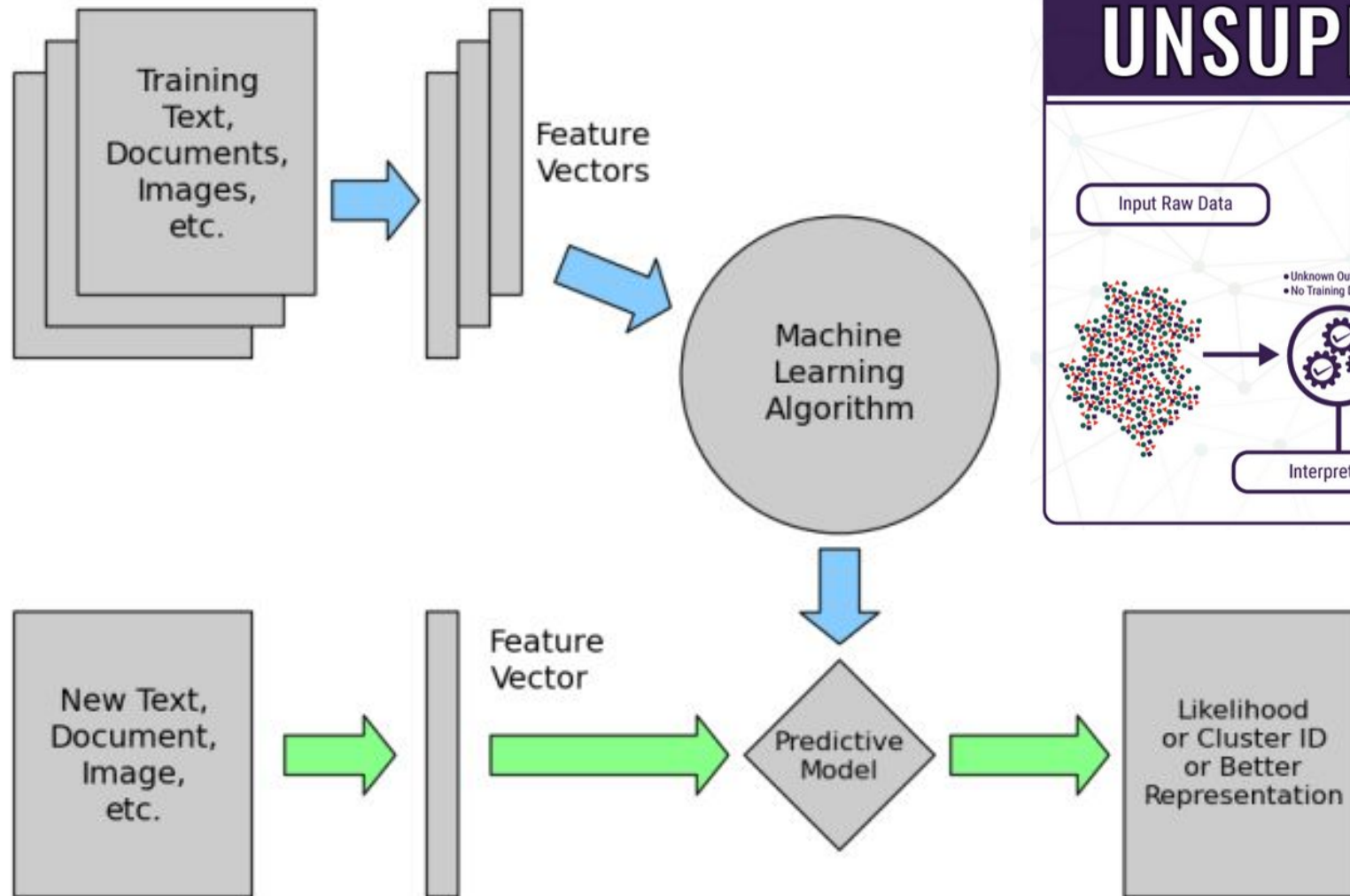
SUPERVISED LEARNING



Supervised Learning

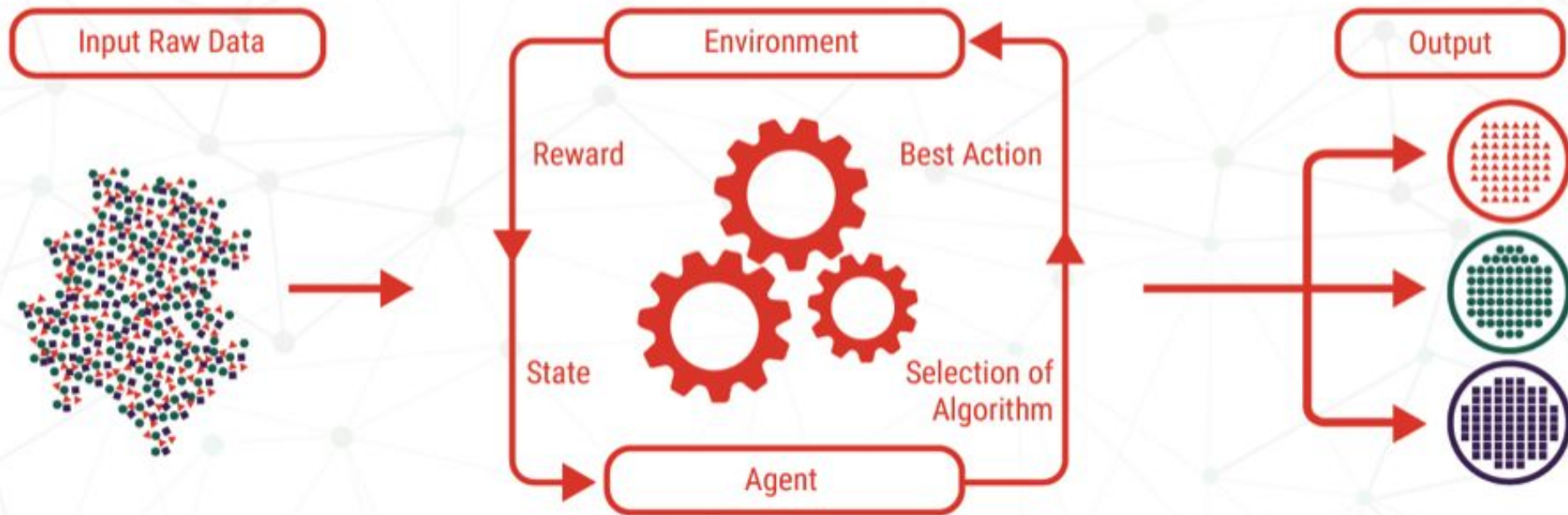


Unsupervised Learning

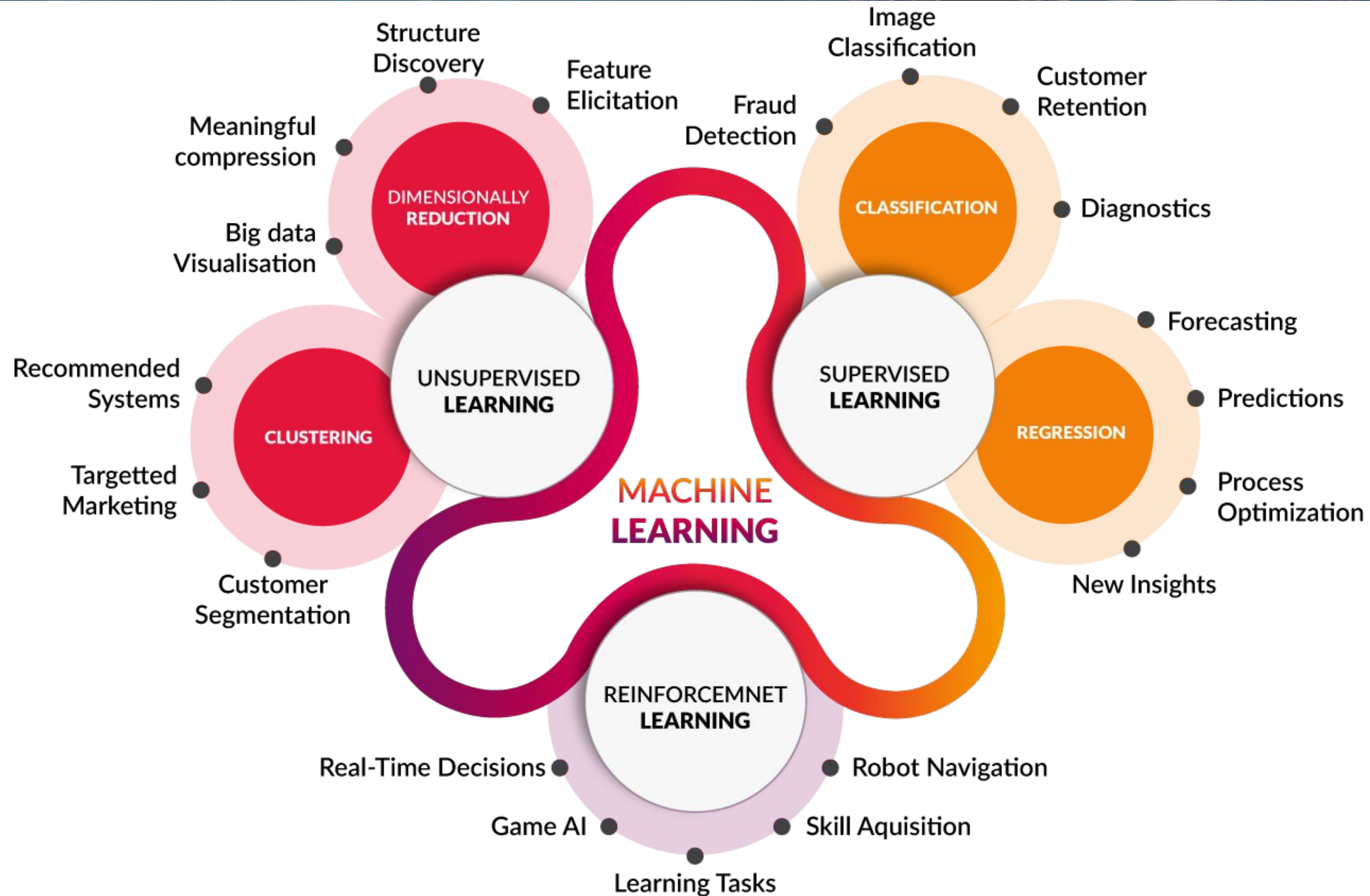


Reinforcement Learning

REINFORCEMENT LEARNING



Machine Learning Extended





Q & A



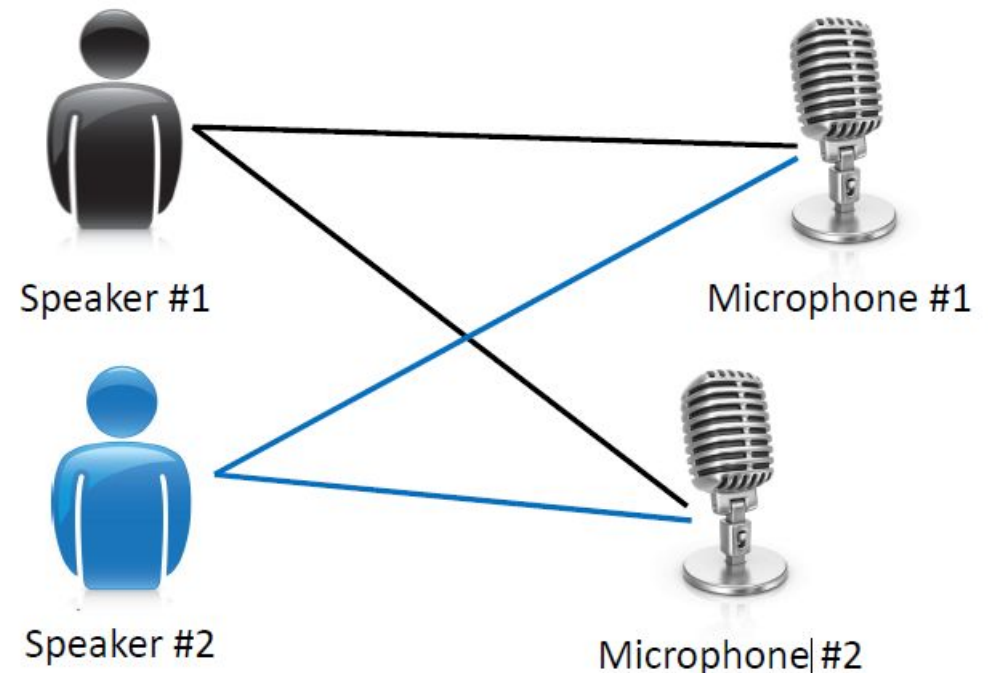
Clustering / Classification / Regression

Clustering / Classification / Regression ??

- Given data about the size of houses on the real estate market, try to predict their price.
- Given a picture of a person, we have to predict their age on the basis of the given picture
- Given a patient with a tumor, we have to predict whether the tumor is malignant or benign.
- Building a system that predicts whether the house "sells for more or less than the asking price."
- Take a collection of 1,000,000 different genes, and find a way to automatically group these genes into groups that are somehow similar or related by different variables, such as lifespan, location, roles, and so on.

- The "Cocktail Party Algorithm", allows you to find structure in a chaotic environment. (i.e. identifying individual voices and music from a mesh of sounds at a [cocktail party](#)).

Cocktail party problem



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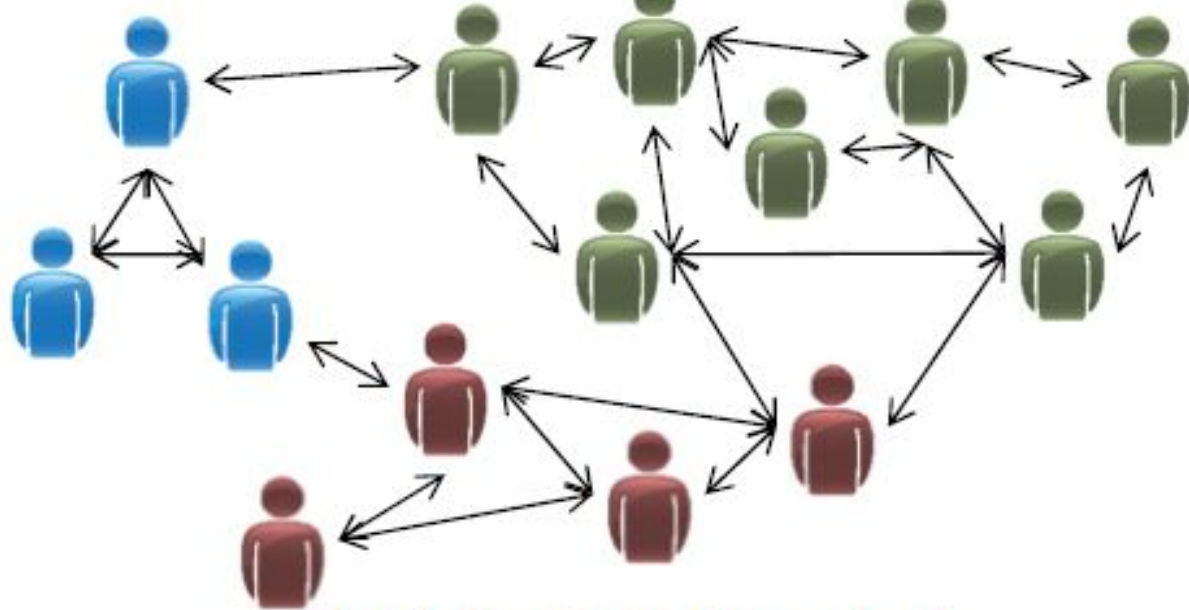




Organize computing clusters



Market segmentation



Social network analysis



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Astronomical data analysis